

MESIN-C Oral Suspension

ANTHELMINTIC (Systemic)

Each 5 mL contains:
Albendazole 200 mg

PRODUCT DESCRIPTION:

Yellow suspension with pineapple and peach flavor

MECHANISM OF ACTION:

Pharmacology

Vermicidal; Albendazole causes degenerative alterations in the tegument and intestinal cells of the worm by binding to the colchicine-sensitive site of tubulin, thus inhibiting its polymerization or assembly into microtubules. The loss of the cytoplasmic microtubules leads to impaired uptake of glucose by the larval and adult stages of the susceptible parasites, and depletes their glycogen stores. Degenerative changes in the endoplasmic reticulum, the mitochondria of the germinal layer, and the subsequent release of lysosomes result in decreased production of adenosine triphosphate (ATP), which is the energy required for the survival of the parasite. Due to diminished energy production, the parasite is immobilized and eventually dies.

Albendazole also has been shown to inhibit the enzyme fumarate reductase, which is helminth-specific. This action may be considered secondary to the effect on the microtubules due to decreased absorption of glucose. This action occurs in the presence of reduced amounts of nicotinamide-adenine dinucleotide in reduced form (NADH), which is a coenzyme involved in many cellular oxidation-reduction reactions.

Albendazole has larvicidal effects in necatoriasis and ovidical effects in ascariasis, ancylostomiasis, and trichuriasis.

Pharmacokinetics

Absorption: Poorly and erratically absorbed from the gastrointestinal tract. Absorption of Albendazole is greatly increased if the medication is taken with food containing relatively high concentrations of fat.

Distribution: Widely distributed throughout the body into bile, cerebrospinal fluid (CSF), fluid in hydatid cysts, liver, serum, and urine. Protein binding: High (70%)

Biotransformation: Hepatic; rapidly and extensively metabolized mainly to the active metabolite, Albendazole sulfoxide, which appears in the systemic circulation at detectable concentrations. Albendazole is also metabolized to the 6-hydroxy sulfoxide and sulfone metabolites, but not in sufficient quantities to be detected consistently in the plasma.

Half-life: Single dose- 400mg: Approximately 8 to 12 hours; 15 mg/ Kg: Approximately 10 to 15 hours

Time to peak concentration: Single dose- 400 mg: Approximately 2 to 5 hours; 15 mg/ Kg: Approximately 4 hours

Peak serum concentration: Single dose- 400 mg: Approximately 0.46 to 1.58 mcg/ mL; 15 mg/ Kg: Approximately 0.45 to 2.96 mcg/ mL

Elimination: Renal- Approximately 1% is excreted in the urine as Albendazole sulfoxide during the first 24 hours. Other metabolites also are renally excreted; Fecal: A small amount is found in the feces

INDICATION:

Used in the treatment of Hydatid disease; Neurocysticercosis; Ascariasis; Capillariasis; Cutaneous larva migrans; Enterobiasis; Hookworm infections; Strongyloidiasis; Taeniasis; Trichostrongylidiasis; Trichuriasis

DOSAGE AND ADMINISTRATION:

Oral

Usual Adult and Adolescent Dose

Ascariasis; Enterobiasis; Hookworm infections; Trichuriasis: 400 mg as a single dose for one day. Treatment may be repeated in three weeks.

Capillariasis: 200 mg two times a day for ten days.

Cutaneous larva migrans: 400 mg once a day for three days.

Hydatid disease: 800 mg per day for twenty-eight days. Treatment may be repeated as necessary. Up to two or three cycles of Albendazole treatment may be given. For inoperable hydatid cysts, up to five cycles may be given.

Note: In hydatid disease, the dose of Albendazole for patients under 60 Kg of body weight is 12 mg per Kg of body weight per day.

Neurocysticercosis: 15 mg per Kg of body weight per day for thirty days. Treatment may be repeated as necessary.

Strongyloidiasis or Taeniasis: 400 mg as a single dose per day for three days. Treatment may be repeated in three weeks.

Trichostrongylidiasis: 400 mg as a single dose.

Note: In the treatment of giardiasis, a dose of 400 mg per day for five days has been used.

Usual Pediatric Dose

Ascariasis or Enterobiasis or Hookworm infections or Trichuriasis: Children up to 2 years of age: 200 mg as a single dose for one day. Treatment may be repeated in three weeks; Children 2 years of age and over: See Usual adult and adolescent dose.

Capillariasis or Neurocysticercosis or Trichostrongylidiasis: See Usual adult and adolescent dose.

Cutaneous larva migrans: Children: 5 mg per Kg of body weight per day for three days.

Hydatid disease: Children up to 6 years of age: Dosage has not been established; Children 6 years of age and over: 10 to 15 mg per Kg of body weight per day for twenty eight days. Treatment may be repeated as necessary.

Strongyloidiasis or Taeniasis: Children up to 2 years of age: 200 mg as a single dose per day for three consecutive days.

Treatment may be repeated in three weeks; Children 2 years of age and over: See Usual adult and adolescent dose.

Note: In the treatment of giardiasis, see Usual adult and adolescent dose.

PRECAUTION:

Should not be administered during confirmed or suspected pregnancy.*

Carcinogenicity: Albendazole has been tested in different species of animals and has not been found to be carcinogenic or mutagenic.

Pediatrics: Limited studies on the relationship of age to the effects on the Albendazole have been performed in children up to 6 years of age. Although hydatid disease is uncommon in infants and young children, no pediatric-specific problems have been documented in infants and young children who were treated with Albendazole for hydatid disease.

Geriatrics: Appropriate studies on the relationship of age to the effects of the Albendazole have been performed in the geriatric population. However, no geriatrics-specific problems have been documented to date.

Fertility: Albendazole has not been shown to cause adverse effects on male or female fertility.

Pregnancy: Adequate and well-controlled studies in humans have not been done. However, Albendazole is not recommended in pregnant women because of its teratogenic effects in animals. FDA Pregnancy Category C.

Breast-feeding: Albendazole is distributed into the milk of lactating animals. It is not known whether Albendazole is distributed into human breast milk. However, problems in humans have not been documented.

SIDE EFFECT:

Those indicating need for medical attention:

Incidence more frequent

Abnormal liver function test results

Incidence rare

Hypersensitivity, neutropenia-with high doses, reversible.

CONTRAINDICATION:

Cysticercosis involving the retina; Hepatic function impairment, such as in hepatic cirrhosis.

DRUG INTERACTION:

Cimetidine (Albendazole sulfoxide concentrations are increased in patients treated with Cimetidine; increased concentrations are observed in bile and cystic fluid); Corticosteroids (Concurrent use with Albendazole increases steady state through concentrations of Albendazole sulfoxide); Praziquantel (Concurrent use with Albendazole increases mean plasma concentration and area under the plasma concentration-time curve of Albendazole sulfoxide)

OVERDOSE AND TREATMENT:

Since no specific antidote is known, recommended treatment of Albendazole overdose consists of the following:

To decrease absorption - Gastric lavage may be undertaken within the first 2 to 3 hours after ingestion.

Symptomatic treatment may be given.

Supportive measures such as maintaining an open airway, respiration, and circulation may be instituted. Patients in whom intentional overdose is confirmed or suspected should be referred for psychiatric consultation.

SHELF LIFE:

Two (2) years

STORAGE:

Store at temperature of not more than 30°C.

SUPPLY:

Bottle 10 mL

Manufactured by:
UNISON LABORATORIES CO., LTD.
19 Moo 4, Klong Udonchojorn, Muang Chachoengsao,
Chachoengsao 24000 Thailand

Marketing Authorization Holder/ Importer:

MEDISPEC (M) SDN. BHD.

No. 55 & 57, Lorong Sempadan 2, (Off Boundary Road),

11400 Ayer Itam, Penang, Malaysia

C-MU/251012-03(EG)

(USPDI : Micromedex Healthcare 2006 Series.; * IRA Memo 7-166 (D))

IMML 0021

12.0 cm
29 cm

Print : Fit to Page

Thickness	60 UNITS 11.17 = 0.06 (mm)
Packaging Type	152x27x118mm
Code No.	IMML 0021
Product Name	MESIN-C 200 MG

PM Specification